LOT PAGER ??? 07-PRODUCTION-100

LOT® PAGER — 100-Unit Production Run

1. Build plan

Step	Who	Lead time	Notes
Gerber/BOM freeze	LOT	—	After DVT pass
PCB fab + SMT, 110 pcs	PCBWay	3–4 wk	Consign scarce ICs
CNC 316L, 110 sets	PCBWay CNC / local shop	3–5 wk	+10 % scrap
Polish (back plates)	Shop	1–2 wk	Mirror finish yield ~85 %
Windows (camera, screen, antenna strip)	Glass/sapphire vendor	3 wk	Custom
Battery, coil, displays, cameras	Distributor	1–6 wk	Order early
Final assembly + test	LOT (or EMS)	2 wk	Jig + test script
Pack + Qi pad	LOT	1 wk	

2. Test fixture (per unit)

1. Flash + provision (UART/pogo), burn device ID/QR.
2. Self-test: display colour bars, button, buzzer, sensors read within range, camera frame, Qi charge current, Wi-Fi RSSI through the closed shell.
3. 24 h burn-in at 5 units/shelf; sample 10 for 72 h.
4. Log results per serial in a CSV; fail → rework bin.

3. Compliance (needed before sale, not for internal pilot)

FCC Part 15 (unintentional + intentional radiator via modular ESP32 cert reuse – antenna change may void; budget pre-scan), CE-RED, UN38.3 for the cell, IEC 62368-1, RoHS. Pre-scan $2–4k; full ~$10–20k. Internal-only distribution of 100 units avoids sale but not safety risk (battery).

4. Costs (100 units, +10 % overage)

Per 02-BOM: **$9–15k** parts+fab, **$1.5–4k** NRE, **$2–4k** pre-scan. Total **$12–23k**.

5. Acceptance

Yield ≥ 90 %, 7-day battery claim ≥ 5 days measured, Qi charge 0→90 % ≤ 90 min, notification latency p95 ≤ 30 s, zero thermals > 40 °C, 1 m drop onto hardwood ×6 faces.